

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	MS-ENVE
Program Name	Master of Science in Environmental Engineering
Assessment Year	2024-2025
Program Assessment Coordinator	Brent Stephens
Year of Current Assessment Plan	2024-2025
No Assessment This Year per Plan	☐
LO(s) Assessed	4
Presented at program authority meeting date: <i>Please attach minutes</i>	n/a

Methodology

The Master of Science in Environmental Engineering program is a research and thesis-based graduate degree program oriented toward students who wish to develop more knowledge in multiple areas of environmental engineering. The program provides students with opportunities to gain experience in a variety of environmental engineering domains, including water and wastewater treatment, water resources engineering, indoor and outdoor air quality, hazardous waste management, environmental health and safety and industrial hygiene, and environmental sustainability.

The program is intended for preparation for advanced engineering practice rooted in the principles of physics, chemistry, biology, and mathematics to develop and analyze solutions to environmental problems, including air and water pollution control, water and wastewater treatment systems design, climate change, resource depletion, and more. Critical to each of these application areas is a strong foundation in the fundamental science and engineering principles needed to assess contaminant fate and transport in the environment.

The primary learning objectives for this program are for students to:

1. Understand contaminant fate and transport in the environment.
2. Formulate mass and energy balances and apply modeling concepts to practical environmental science and engineering problems.
3. Assess the sustainability of engineered systems.
4. Formulate research questions, conduct literature reviews, and utilize relevant tools to advance knowledge in environmental engineering

Artifact(s) used to evaluate student learning:

- LOs #1-3: Performance on assignments and exams in one of the required courses in the program (typically ENVE 506 Chemodynamics, ENVE 522 Global Environmental Change and Sustainability Analysis, or ENVE 542 Physicochemical Processes in Environmental Engineering)
- LO #4: Master's thesis (written thesis and oral defense evaluation)

Semester(s) in which artifacts are collected and assessed:

- LOs #1-4: End of each academic year

Sample size:

- LOs #1-3: Each student in the program who took the selected course (ENVE 506, 522, and 542 are typically on rotation and will thus alternate in offering and assessment schedule)
- LO #4: Each student who defended and completed their master's thesis in each academic year

Data to report:

- Results are reported as statistical distributions (i.e., histograms) and summary statistics for the sample of students in each year on each performance indicator

Results

Data table

A total of 5 students were enrolled in the program in Fall 2024, including 2 who were pursuing the degree as a co-terminal student and 3 as a traditional master's student. Among these 5 students, 4 took ENVE 506 in Fall 2024 and are thus used for assessment of LOs #1-3 in this cycle, while 1 completed their master's thesis in this academic year.

The following artifacts were used to assess performance toward each LO:

- LO #1 (Understand contaminant fate and transport in the environment):
 - HW assignments #4 and #7 and the final exam
- LO #2 Formulate mass and energy balances and apply modeling concepts to practical environmental science and engineering problems):
 - HW assignments #1, #2, #3, #5, and #6
- LO #3 (Assess the sustainability of engineered systems):
 - Final report
- LO #4 (Formulate research questions, conduct literature reviews, and utilize relevant tools to advance knowledge in environmental engineering)
 - Master's thesis (written and oral defense)

Table 1 shows results from the 4 students in the program who took ENVE 506 across LOs #1, #2, and #3. Scores range from 0 to 100 on a fractional basis. Definitions of proficiency levels are as follows: ≥ 0.9 (proficiency), 0.7-0.9 (meets expectations), < 0.7 (does not meet expectations). Distributions of proficiency levels are shown in Figure 1. Artifact samples are archived in Canvas.

Table 1. Results from evaluation rubric applied to LOs #1, #2, and #3 for this academic year

student	LO1	LO1 - DMP	LO2	LO2 - DMP	LO3	LO3 - DMP
student 1	0.92	P	0.97	P	0.92	P
student 2	0.25	D	0.49	D	0.00	D
student 3	0.94	P	0.97	P	0.92	P
student 4	0.00	D	0.00	D	0.00	D

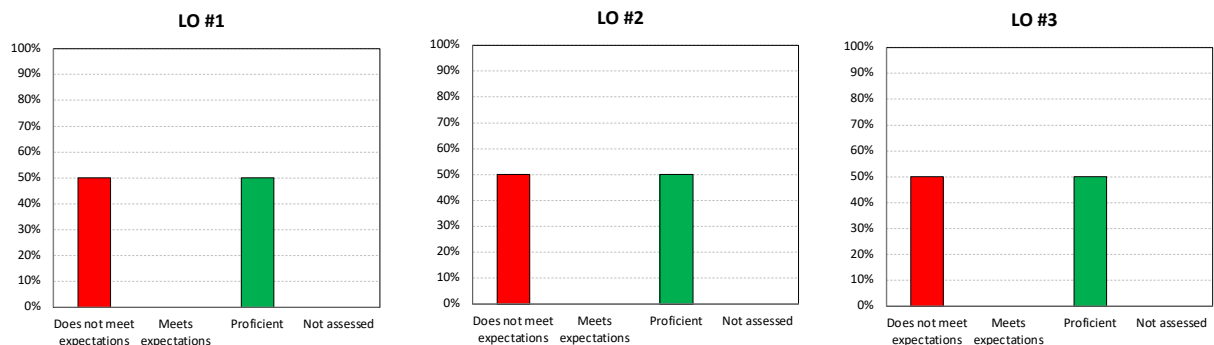


Figure 1. Distribution of proficiency levels for LOs #1, #2, and #3 for this academic year

Regarding LO #4, only one student completed their master's thesis and oral defense in this cycle.

Description of findings

Key findings from this academic year and cohort include:

- Only 2 out of 4 students sampled met or exceeded expectations for LO #1, with both students who were below expectations simply ceasing participation in the course (generally due to personal reasons). Among those 2 students who remained focused in the program, they were both proficient in LO #1.
- Only 2 out of 4 students sampled met or exceeded expectations for LO #2, with both students below expectations effectively ceasing participation in the course due to personal reasons. Otherwise, students demonstrated reasonable proficiency.
- Similar to LO #1, only 2 out of 4 students sampled met or exceeded expectations for LO #3, with both students who were below expectations effectively ceasing participation in the course.
- For LO #4, the one student who completed their master's thesis and oral defense in this cycle passed on the first attempt to defend, suggesting proficiency with no concerns.

Prior CI plan impact

In the previous assessment report, institutional reviewers made several recommendations to improve upon our assessment plans, including converting to an updated format, indicating more clearly which artifacts are used and collected, and to improve upon our CI plans. We had also previously noted that we must continue to keep a close eye on students coming from non-engineering backgrounds. In fact, advisors have attempted to improve preparation by students from non-engineering backgrounds by incorporating stricter prerequisite requirements at admission and we continue to execute these continuous improvement actions. None of the performance shown above raises concerns for background preparation for the program.

This cycle's CI plan

1. In this cycle, we revised our learning objectives to 3 core objectives that match the ME-ENVE program (coursework basis), plus an additional 4th objective that reflects the research-intensive nature of the MS-ENVE program, crafting language that allows for a broader characterization of main goals of the program.

2. We also articulated a clearer plan for assessing the programs LOs #1-3 using more than one course if needed, aiming to assess via two of the more frequently offered courses in the program that are also listed in the required courses list for the program (expanding from ENVE 506 only to include ENVE 506, ENVE 522, or ENVE 542). In this cycle, we had offered ENVE 506 with enrollment from a representative cross-section of the program, so we again used ENVE 506.
3. Continuing with last year's plan, advisors will attempt to improve preparation by students from non-engineering backgrounds by incorporating stricter prerequisite requirements at admission. We are paying closer attention to transcripts in the application review process, including scrutinizing what "engineering" means at different universities across the globe. In these cases, we now accept with deficiency course requirements of CAE 302 Fluid Mechanics, plus any required engineering mathematics courses that are missing.
 - a. One of the students in this cohort needed these prerequisite deficiency courses, and ultimately did not perform well on LOs #1-3, but it is difficult to ascertain whether the challenges were due to preparation or personal problems. Inspection of performance in other courses suggests that the student was able to do well in some courses but not in this most challenging, most totally representative course.
4. Performance toward learning goals below expectations for 2 of the 4 students was particularly concerning in this cycle, although both effectively dropped out of their program pursuit due to personal problems, which makes it difficult to learn much more from their performance.
 - a. Another student who did well towards LOs #1-3 never made substantive progress on their thesis research and ultimately transitioned to the coursework-only ME-ENVE program and graduated without producing a thesis. This student will thus not be assessed on LO #4 given their program change.
5. Finally, we have developed a master's thesis defense evaluation rubric to use for future assessments of LO #4 in the program, and will begin utilizing that tool in the next cycle.